

# Computer Science & IT

## Theory of Computation



Regular Languages

No 1



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# Topics to be Covered



Topic

Regular languages

Topic

Topic

RAVINDRABABU RAVULA

RBR  
 BTech III → Gate - 86  
 BTech IV → Gate - 13  
 ME, IISc (2008)

Refresh

Basics → Easy → medium → Hard



Symbols  $\rightarrow (a, b, c, \dots, z, 0, 1, 2, 3, \dots, 9, \text{男}, \text{女})$

Alphabet  $\rightarrow \Sigma = \{a, b\} \rightarrow$  finite set of symbols

string  $\rightarrow$  sequence of symbols

TOL  $\rightarrow$  CD

Symbols  
↓  
 $\Sigma$   
↓  
Strings

$\Sigma = \{a, b\}$

Strings  $\rightarrow a, b, aab, aba, ab, ba, aa, bb, \dots$   
 $\rightarrow$  every string is always finite



$$\Sigma = \{a, b\}$$

$\hookrightarrow$  The no of strings possible are  $\infty$   
 \* Every string is a finite sequence of symbols

Language  $\rightarrow$  Set of strings  $\begin{cases} \text{finite} \\ \text{infinite} \end{cases}$   
 $\Sigma = \{a, b\}$   
 $L_1 = \{\underline{a}, \underline{b}\} \rightarrow \text{finite}$      $L_2 = \{a, ab, bbb\} \rightarrow \text{finite language}$



$$\Sigma = \{a, b\}$$

$L_3 =$  Set of all strings starting with 'a'

$$= \{a, aa, ab, aaa, aab, \dots\} \rightarrow \text{Infinite language}$$

$10^{10}$  This a  $\infty$  language  
which contains finite strings

$$\Sigma = \{a, b\}$$

$L_4 =$  Set of all strings ending with 'a'

$\{\underline{a}, b\underline{a}, a\underline{a}, ba\underline{a}, \dots\} \rightarrow \infty$  language  
of finite strings



$$a \cdot b = ab$$

$$b \cdot a = ba$$

$$\Sigma = \{a, b\}$$

$$\Sigma \cdot \Sigma = \Sigma^2 = \{a, b\} \cdot \{a, b\} = \{aa, ab, ba, bb\}$$

concatenation

$$\Sigma \cdot \Sigma \cdot \Sigma = \Sigma^3 = \{aaa, aba, abb, \dots, bbb\}$$

$$\Sigma^2 = \text{set of all strings of length 2}$$

$$\Sigma^3 = \text{set of all strings of length 3}$$

$\Sigma^n$  = Set of all strings of length 'n'

$\epsilon$  is a string of length '0'

$$|\epsilon| = 0$$



$$\Sigma = \{a, b\}$$

\*  $\rightarrow$  Kleen closure

aaba

<sup>x</sup> ab...ba

$$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \dots \cup \Sigma^n \dots$$

$\Sigma^* = \{\epsilon, a, b, aa, ab, ba, bb, aaa, \dots\}$

$\Sigma^* \rightarrow$  universal language  $\rightarrow$  Set of all strings possible  
 $\rightarrow \infty$  language of finite string

$\{\}$   $\rightarrow$  empty set  $\rightarrow \phi$   
 $\{\epsilon\}$   $\rightarrow$  set containing one string ' $\epsilon$ '  
 $\phi \neq \{\epsilon\}$   $\epsilon \rightarrow$  null string

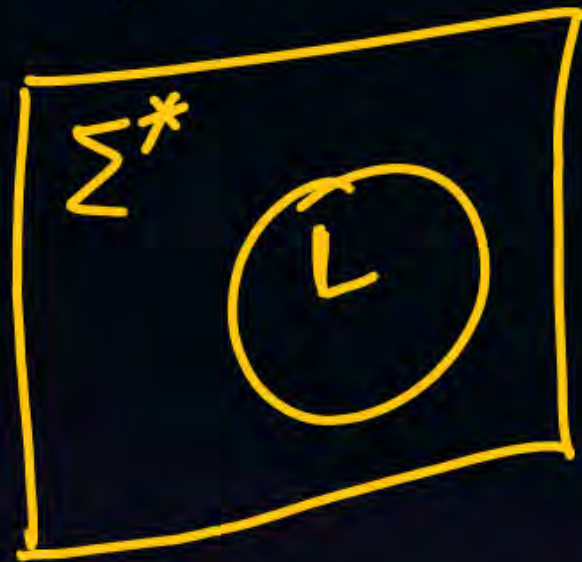


$$|\{\}_{\uparrow}| = 0 \quad |\{\epsilon\}_{\uparrow}| = 1$$

$\Sigma^*$  → universal language

every language is a subset of  $\Sigma^*$

$$\phi = \{\}_{\downarrow} \quad \epsilon \rightarrow \text{string}$$



$\{\}_{\uparrow}$  → Cardinality = 0

$\{\epsilon\}_{\downarrow}$   
Set containing one string of length '0'



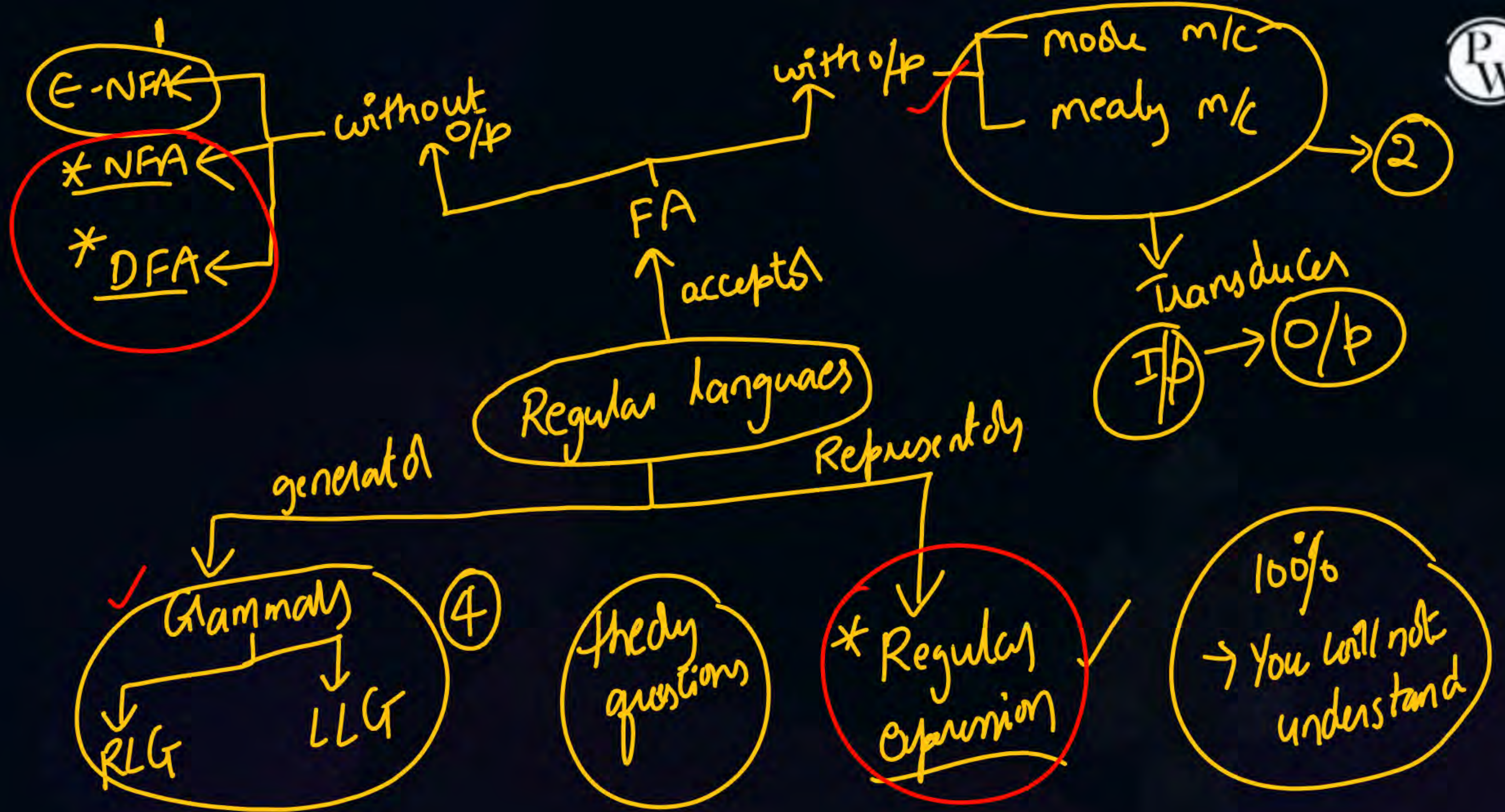


Reg → Reg

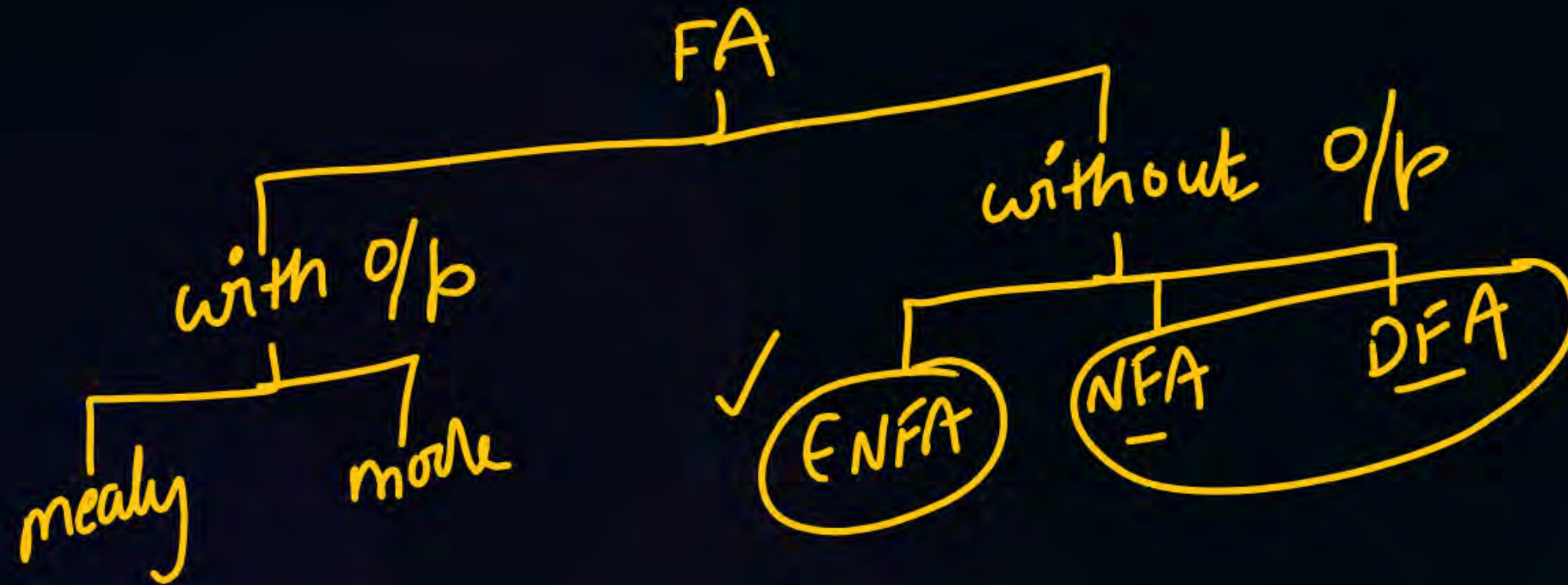


70% of syllabus → Reg languages  
-80%

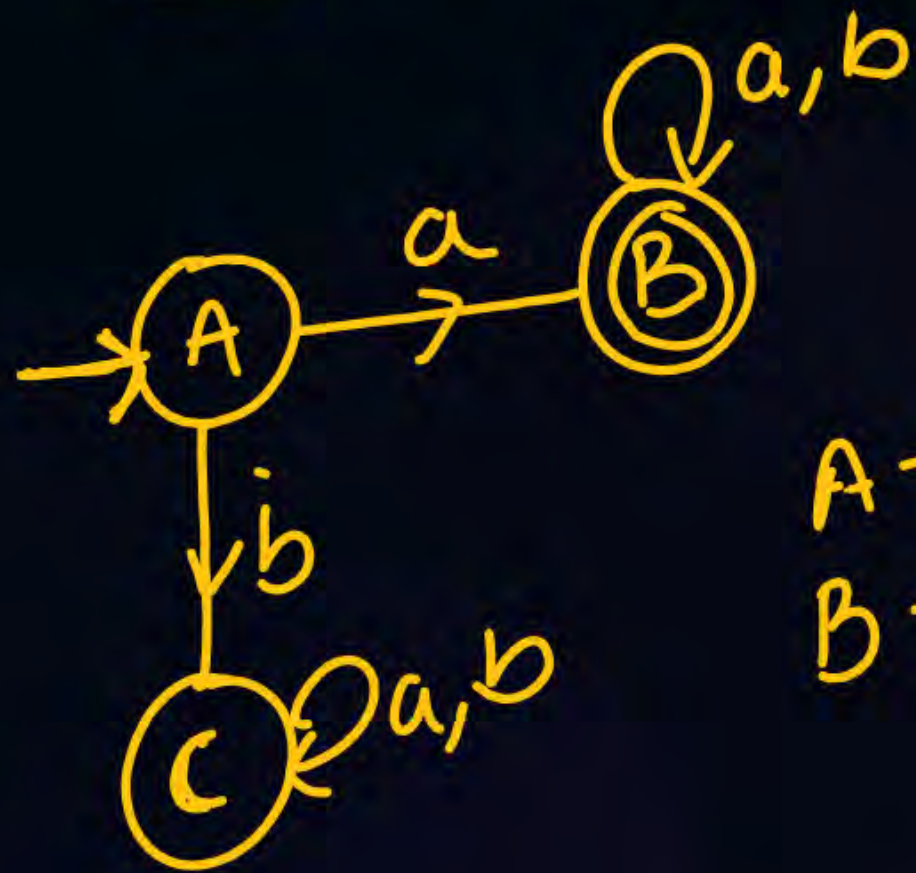








DFA - Deterministic finite automata  
 $\checkmark (\underline{Q}, \underline{\Sigma}, \underline{q_0}, \underline{F})$



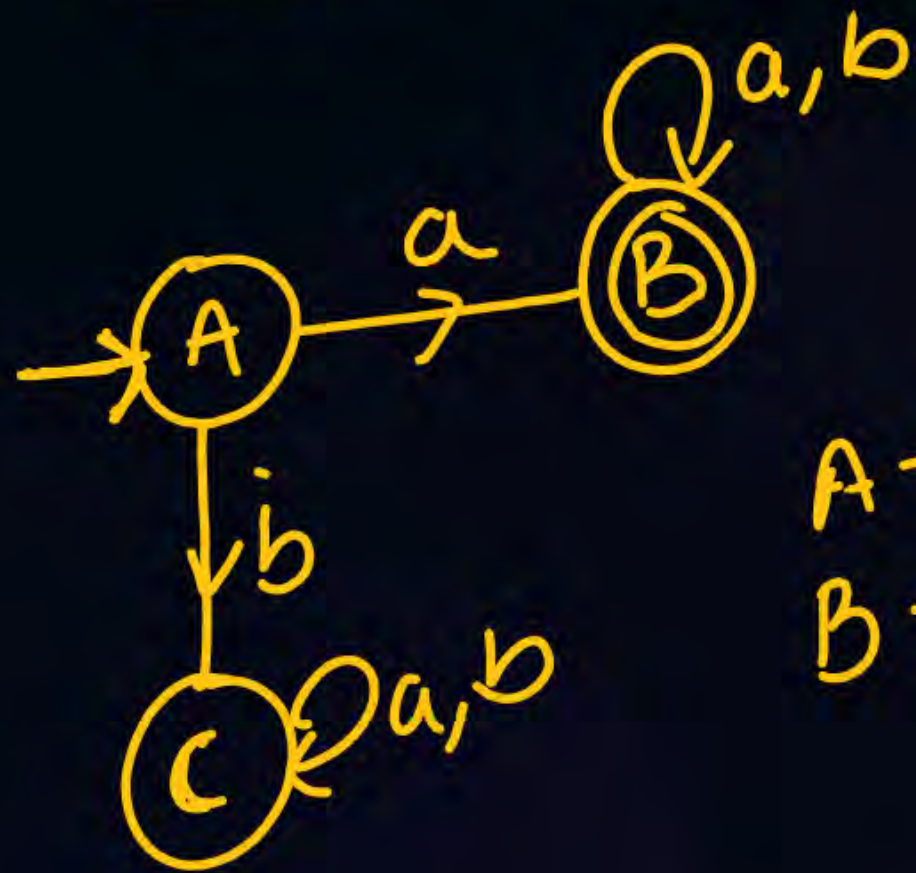
A → start state  
 B → final state

$Q \rightarrow$  finite set of states  
 $\{A, B, C\}$

$\Sigma \rightarrow$  i/p alphabet  
 $\{a, b\}$  → finite



DFA - Deterministic finite automata  
 $\checkmark (Q, \Sigma, \delta, q_0, F)$



A → start state  
 B → final state

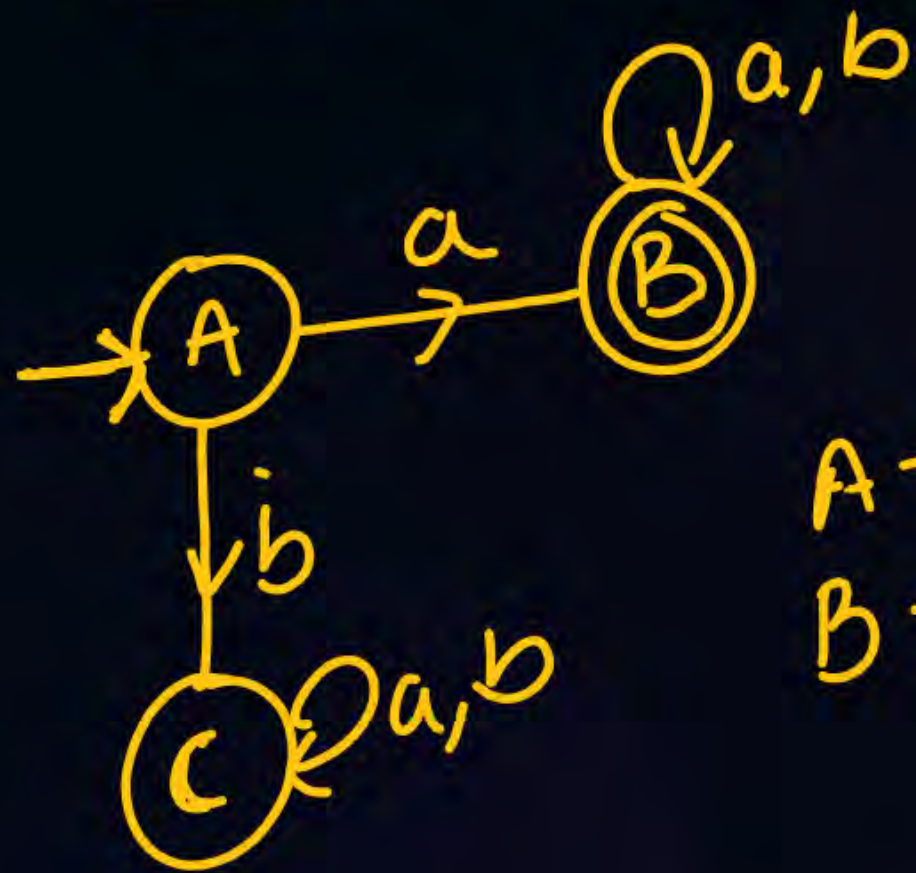
$q_0 \rightarrow$  start state

$q_0 = \underline{A}$

for any machine  
 there will be only  
 one initial state



DFA - Deterministic finite automata  
 $\checkmark (Q, \Sigma, \delta, q_0, F)$



A → start state  
 B → final state

$q_0 \rightarrow$  start state

$F \rightarrow$  set of final states

$F = \{B\} \rightarrow$  For any machine,  
 there can be any no of  
 final states



DFA - Deterministic finite automata

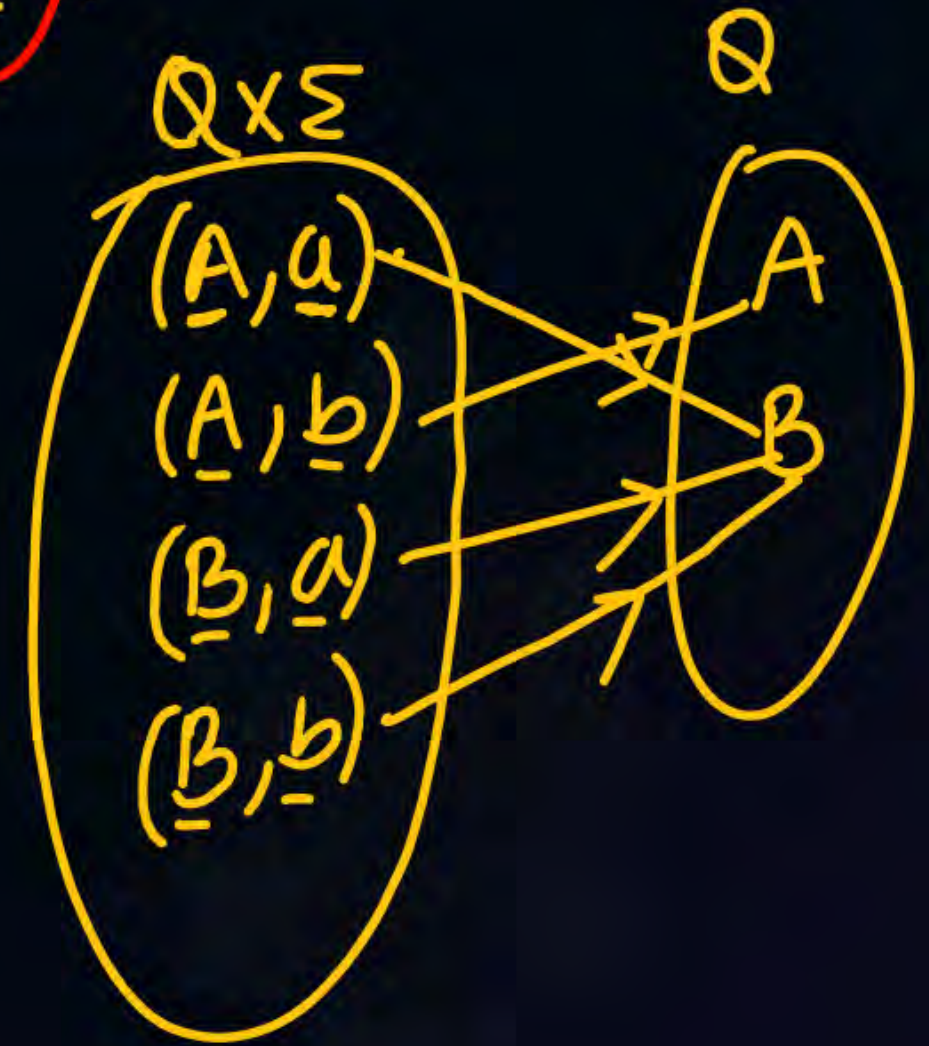
$\checkmark (\underline{Q}, \underline{\Sigma}, \underline{\delta}, \underline{q_0}, \underline{F})$

$\delta: \underline{Q} \times \underline{\Sigma} \rightarrow \underline{Q}$



A → start state  
B → final state

$Q = \{A, B\}$   
 $\Sigma = \{a, b\}$





DFA - Deterministic finite automata

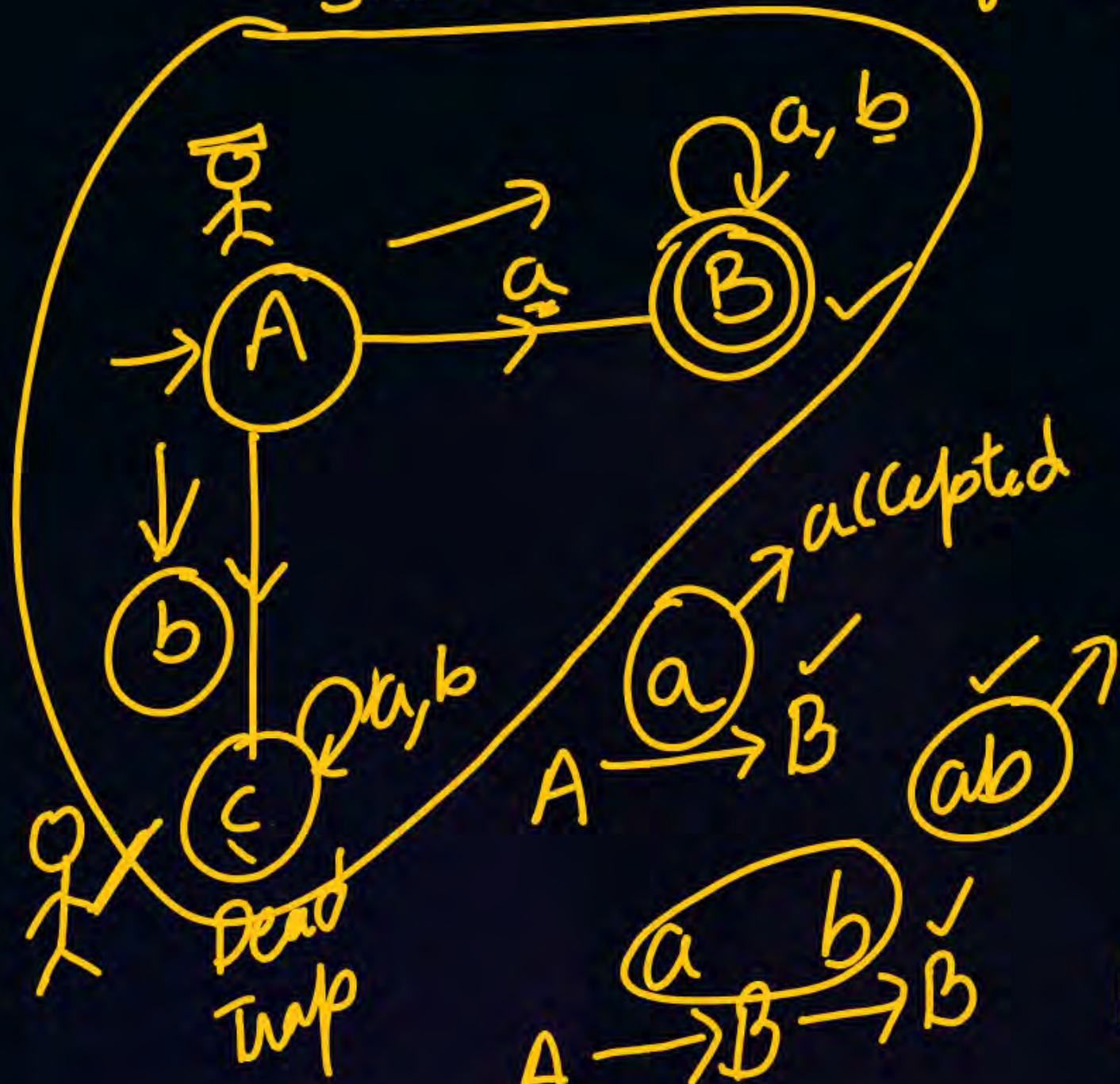
✓ ( $Q$ ,  $\Sigma$ ,  $\delta$ ,  $q_0$ ,  $F$ )  $\delta: \underline{Q} \times \underline{\Sigma} \rightarrow \underline{Q}$



In a DFA, For every state, there will be exactly one transition for every i/p symbol.

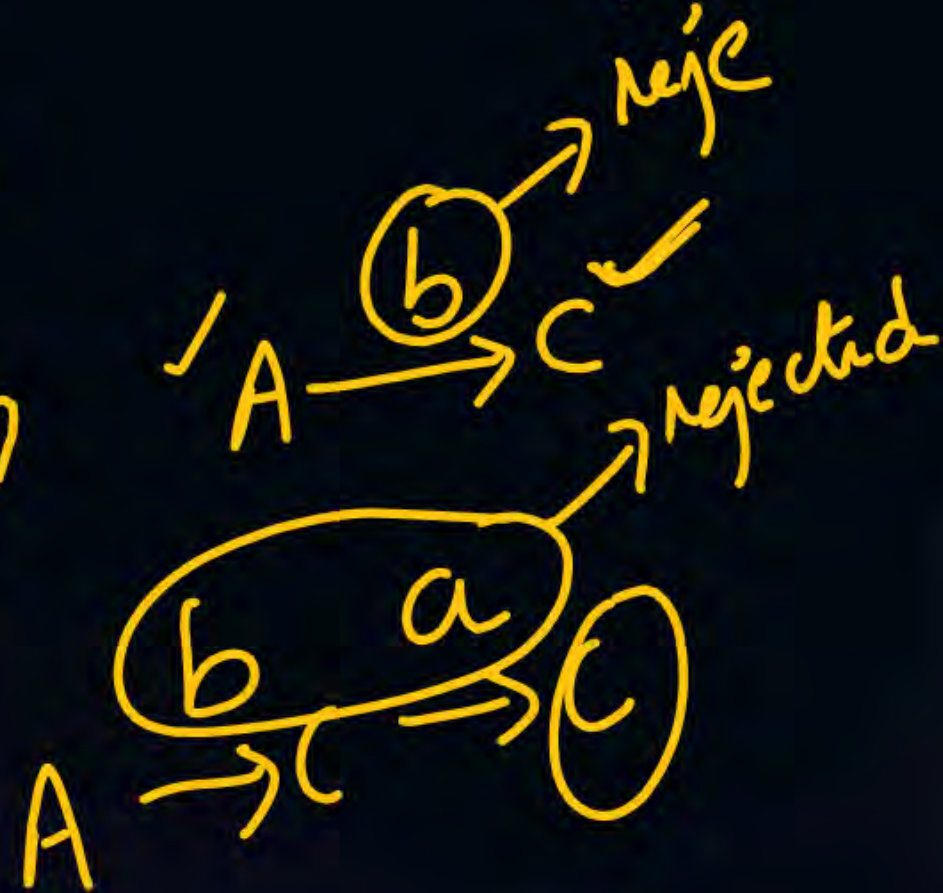


# state transition diagram



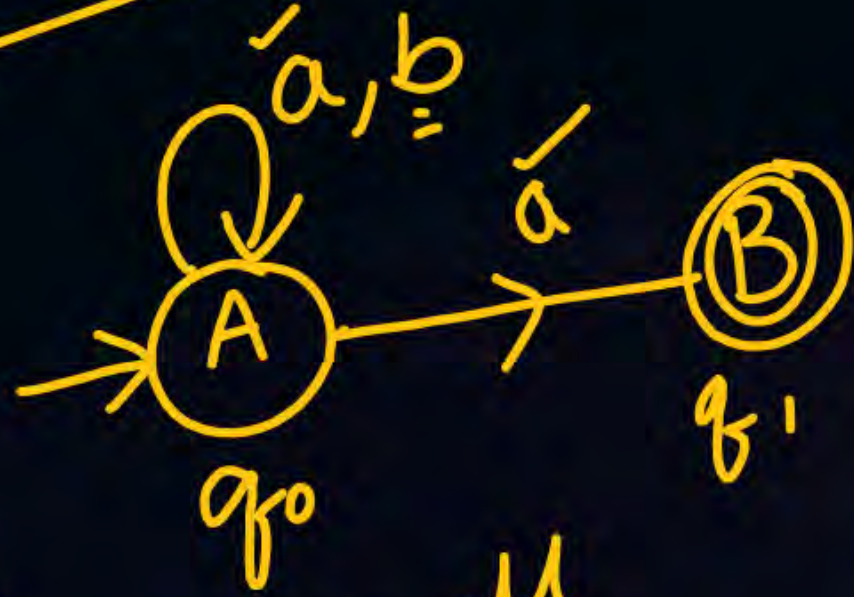
## State transition table

|    | a | b |
|----|---|---|
| →A | B | C |
| *B | B | B |
| C  | C | C |

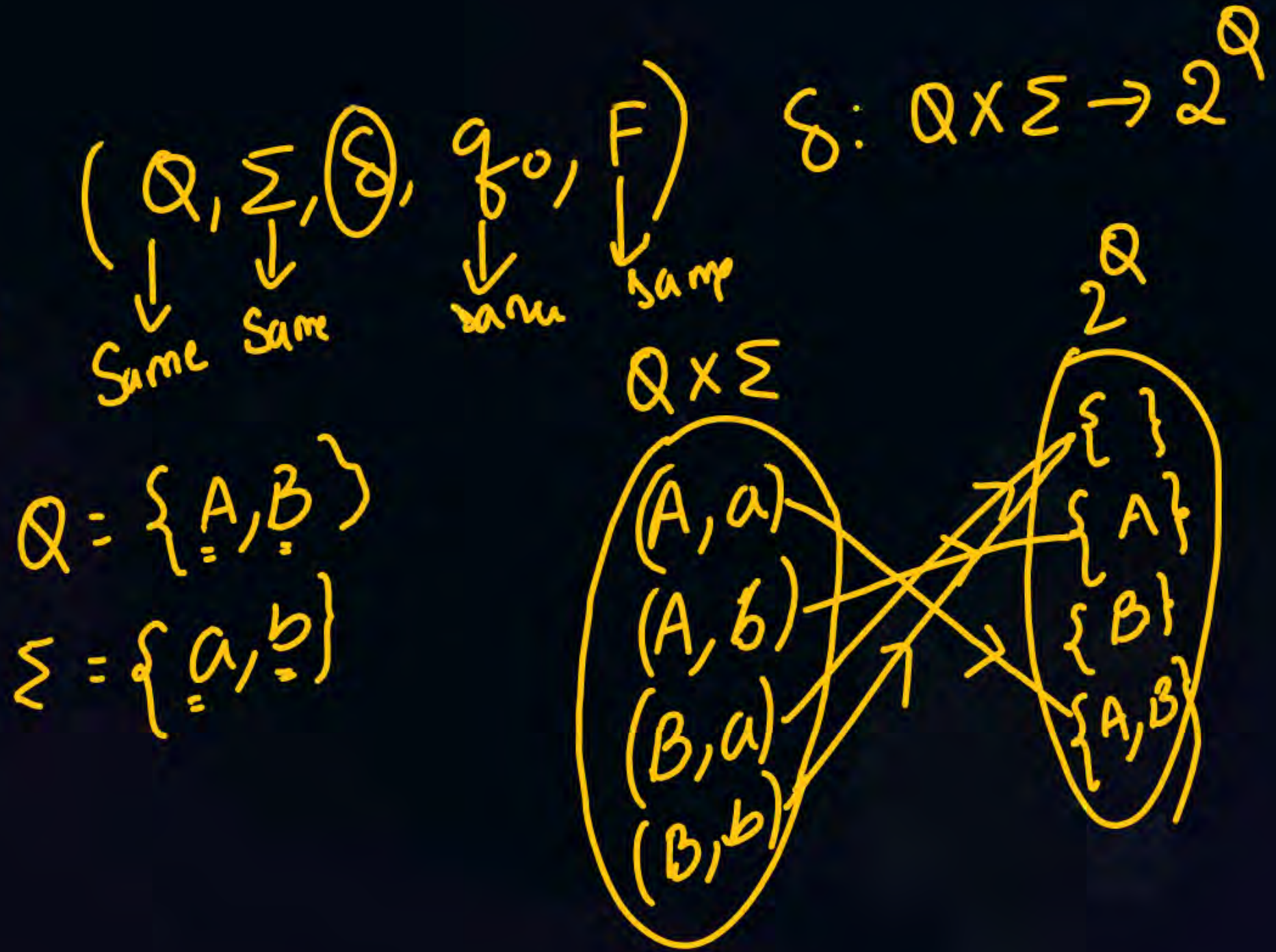




NFA:-

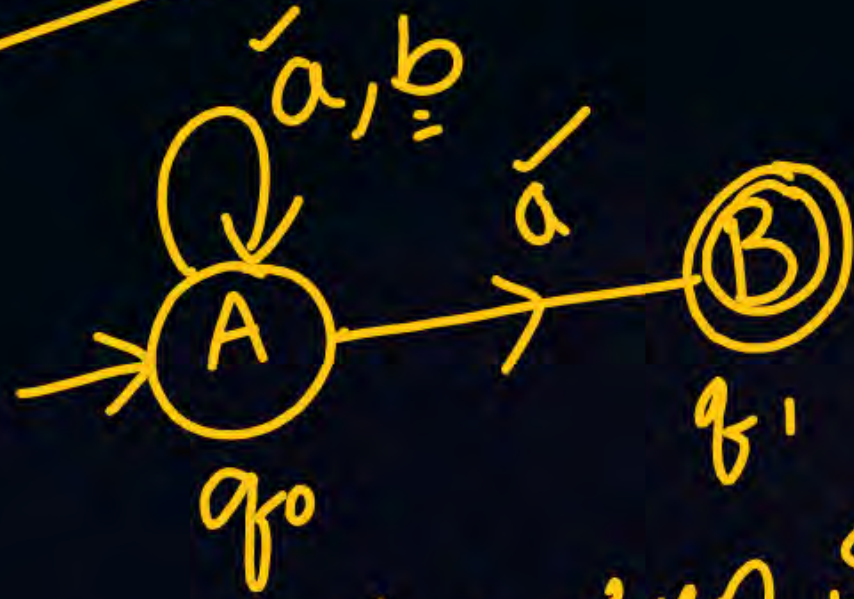


$2^Q \rightarrow$  set of all  
Subsets of Q



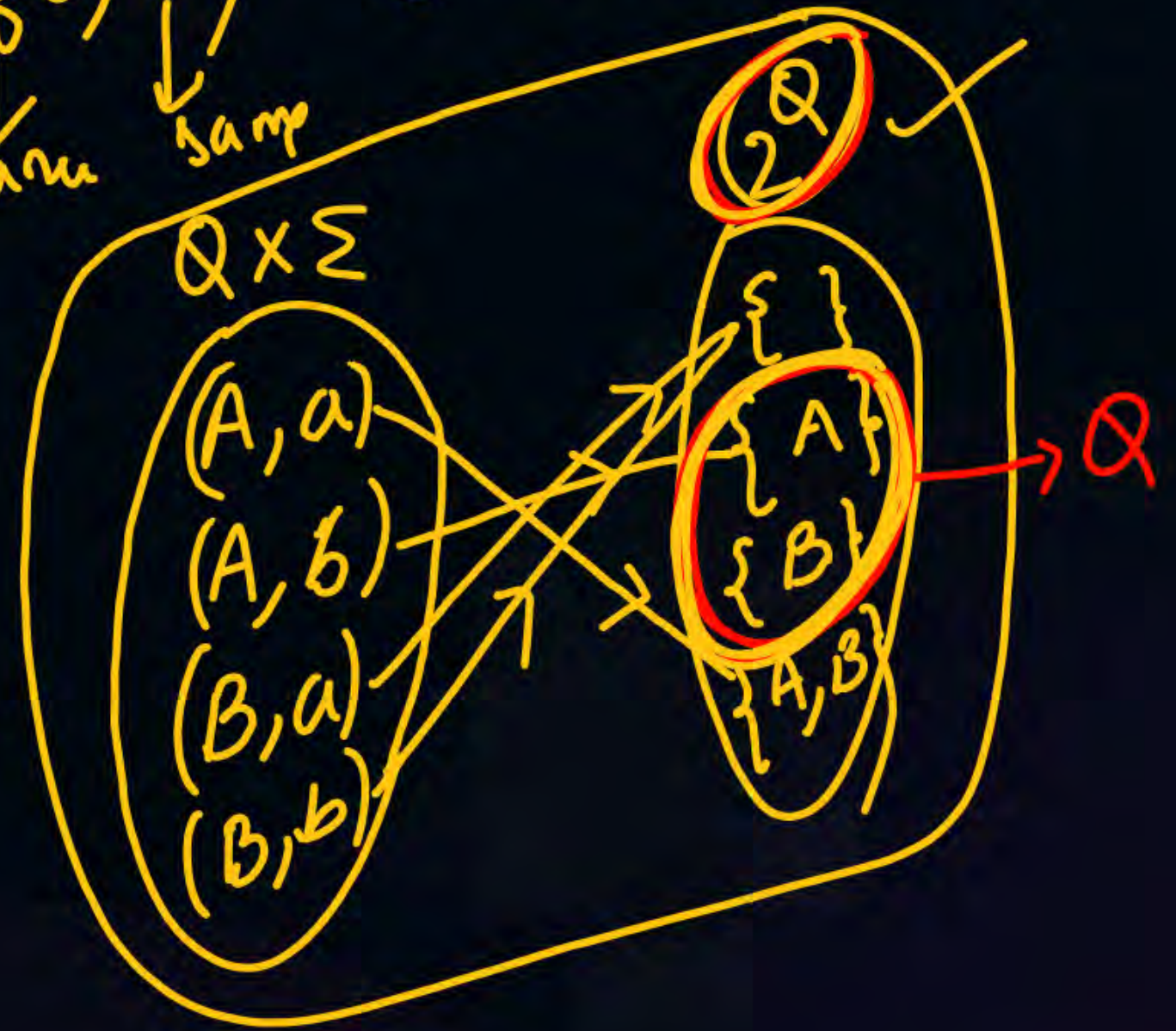


NFA:-



$(Q, \Sigma, \delta, q_0, F)$   
 $\downarrow$  Same  $\downarrow$  Same  $\downarrow$  Same  $\downarrow$  Same

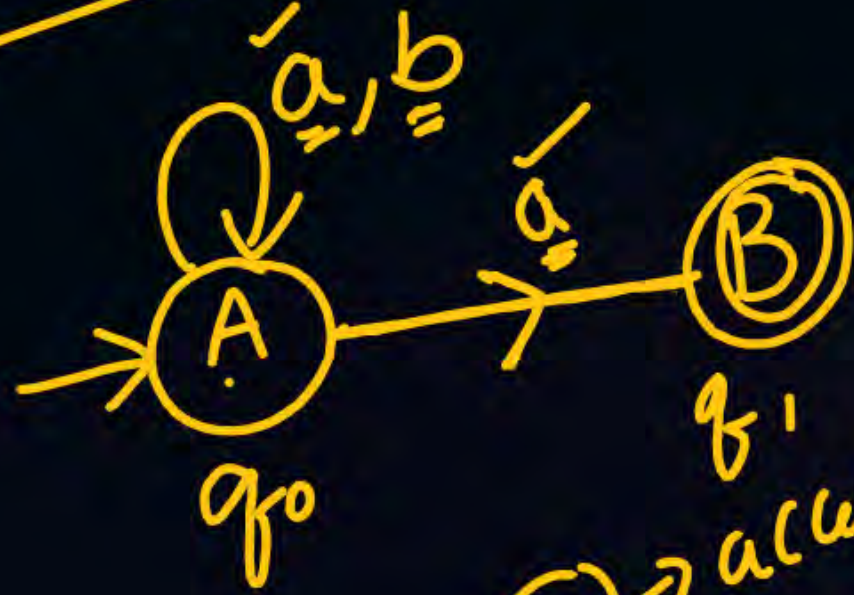
$$\delta: Q \times \Sigma \rightarrow 2^Q$$



NFA: For a given state, on an  
 i/p, NFA can go to 0 states, 1 state  
 or more than one state

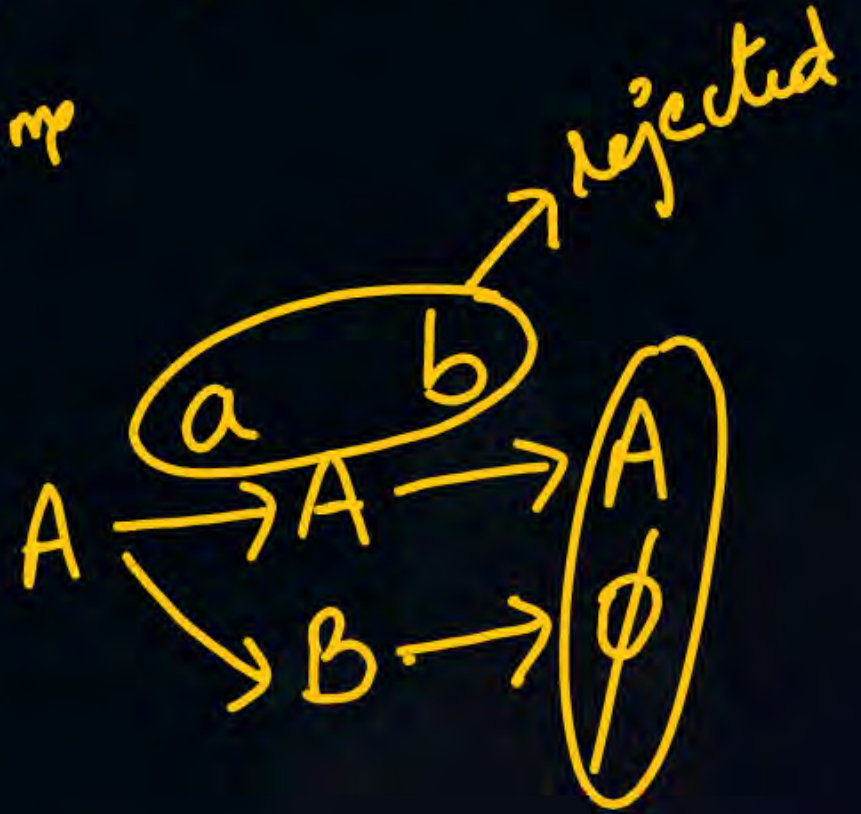
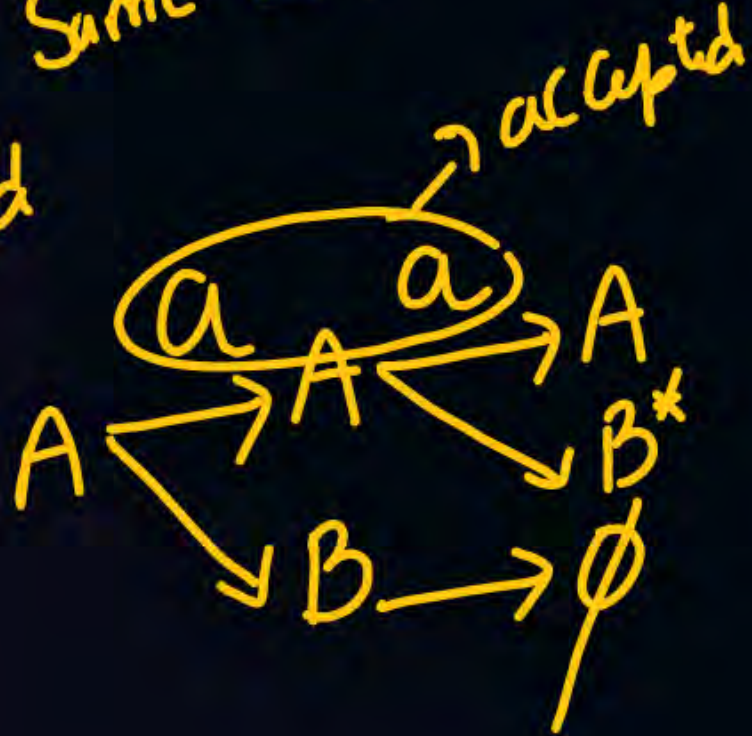
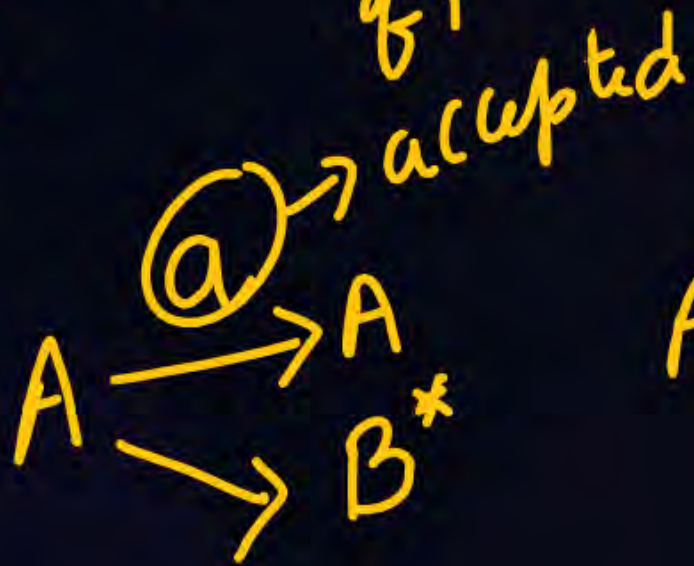


NFA:-



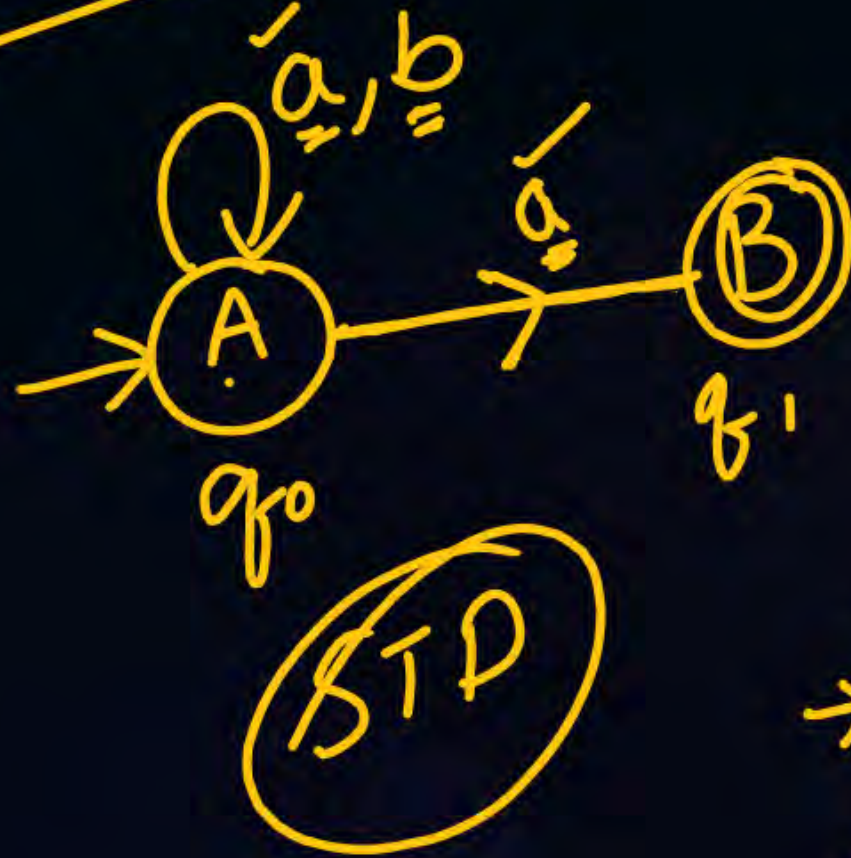
$(Q, \Sigma, \delta, q_0, F)$   
 $\downarrow$  Same  $\downarrow$  Same  $\downarrow$   $q_0$   $\downarrow$   $F$   
 Same Same Same

$\delta: Q \times \Sigma \rightarrow 2^Q$





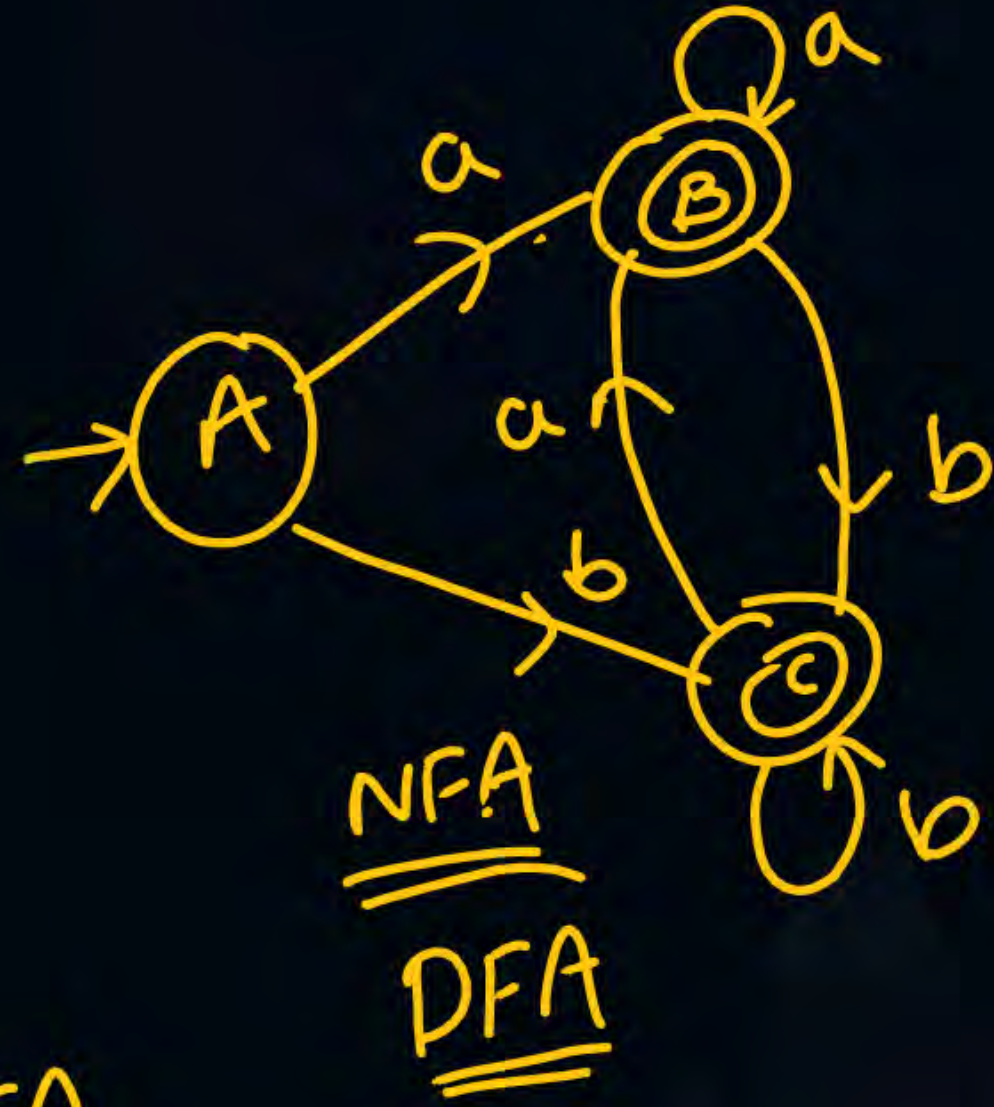
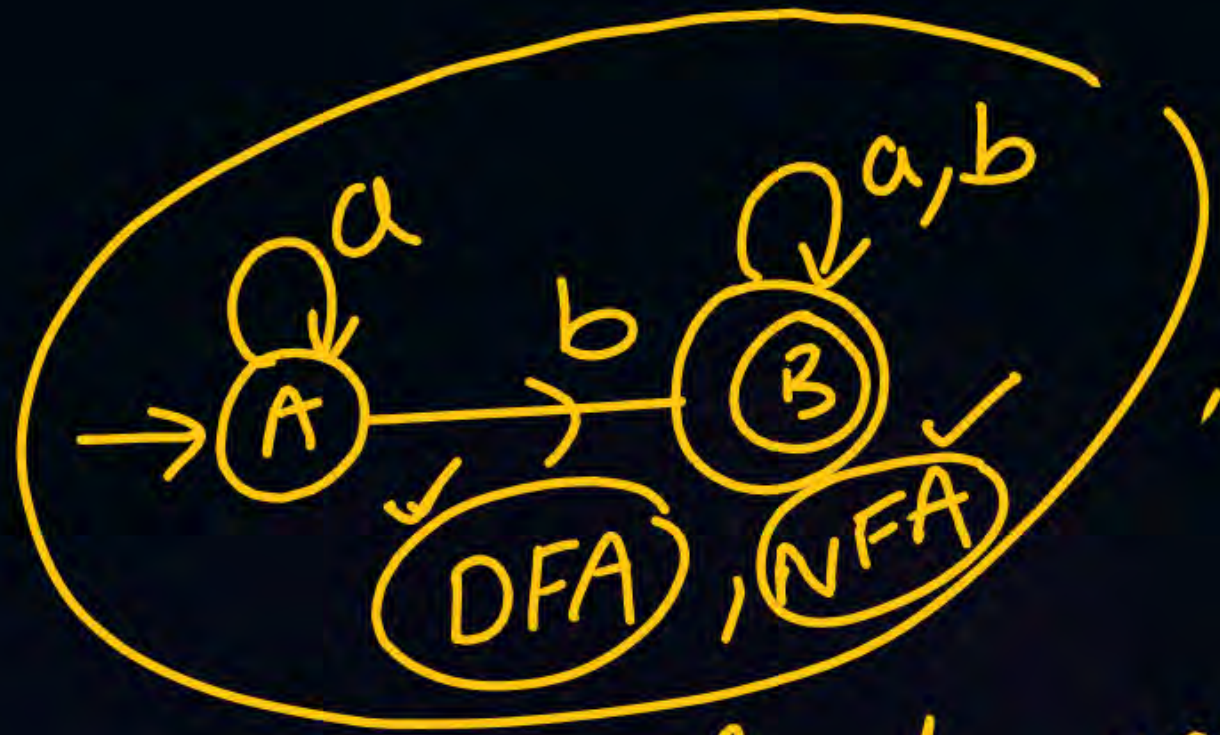
NFA:-



$(Q, \Sigma, \delta, q_0, F)$   
 $\downarrow$  Same  $\downarrow$  Same  $\downarrow$   $\downarrow$   $\downarrow$  Same  
 Same CTI

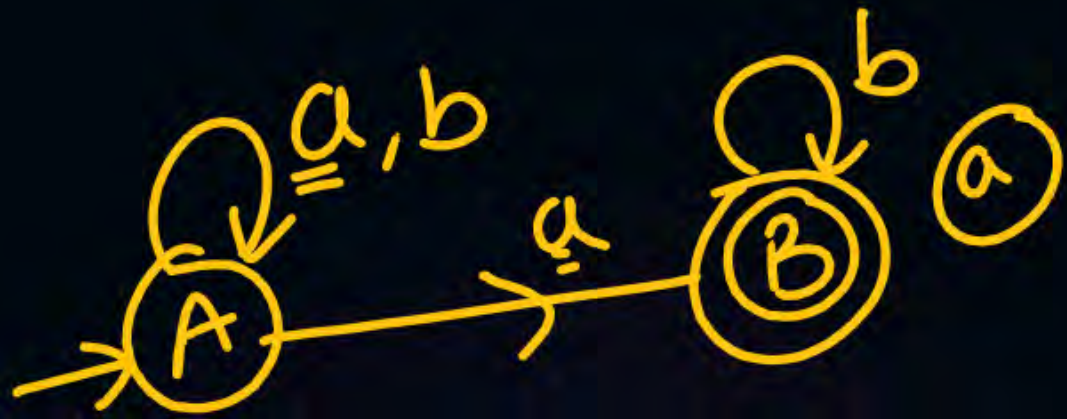
$$\zeta: \mathbb{Q} \times \Sigma \rightarrow 2^{\mathbb{Q}}$$

|     | a      | b      |
|-----|--------|--------|
| → A | {A, B} | {A}    |
| * B | $\phi$ | $\phi$ |



Every DFA is also an NFA  
 Every NFA need not be a DFA

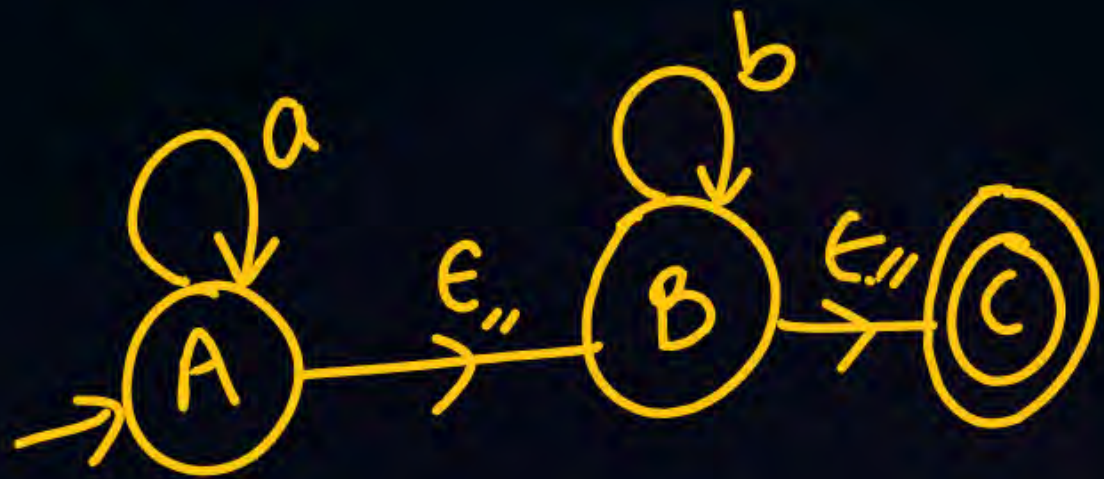




NFA

DFA? X

E-NFA:

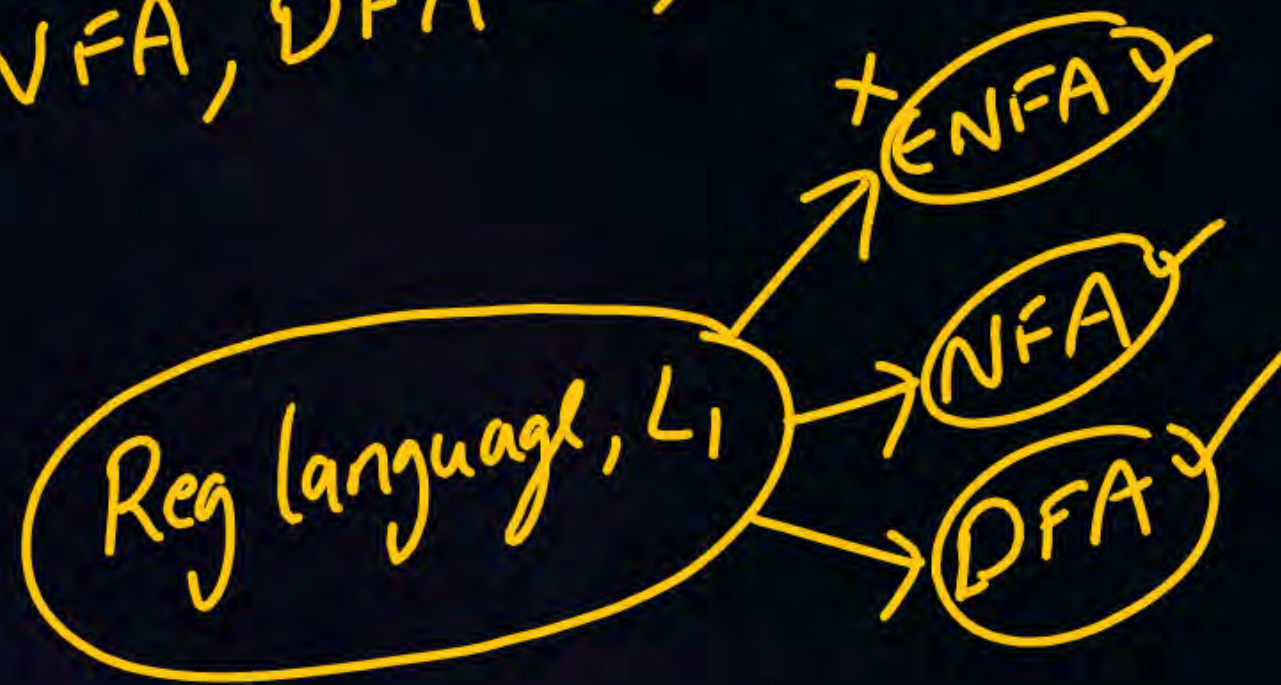


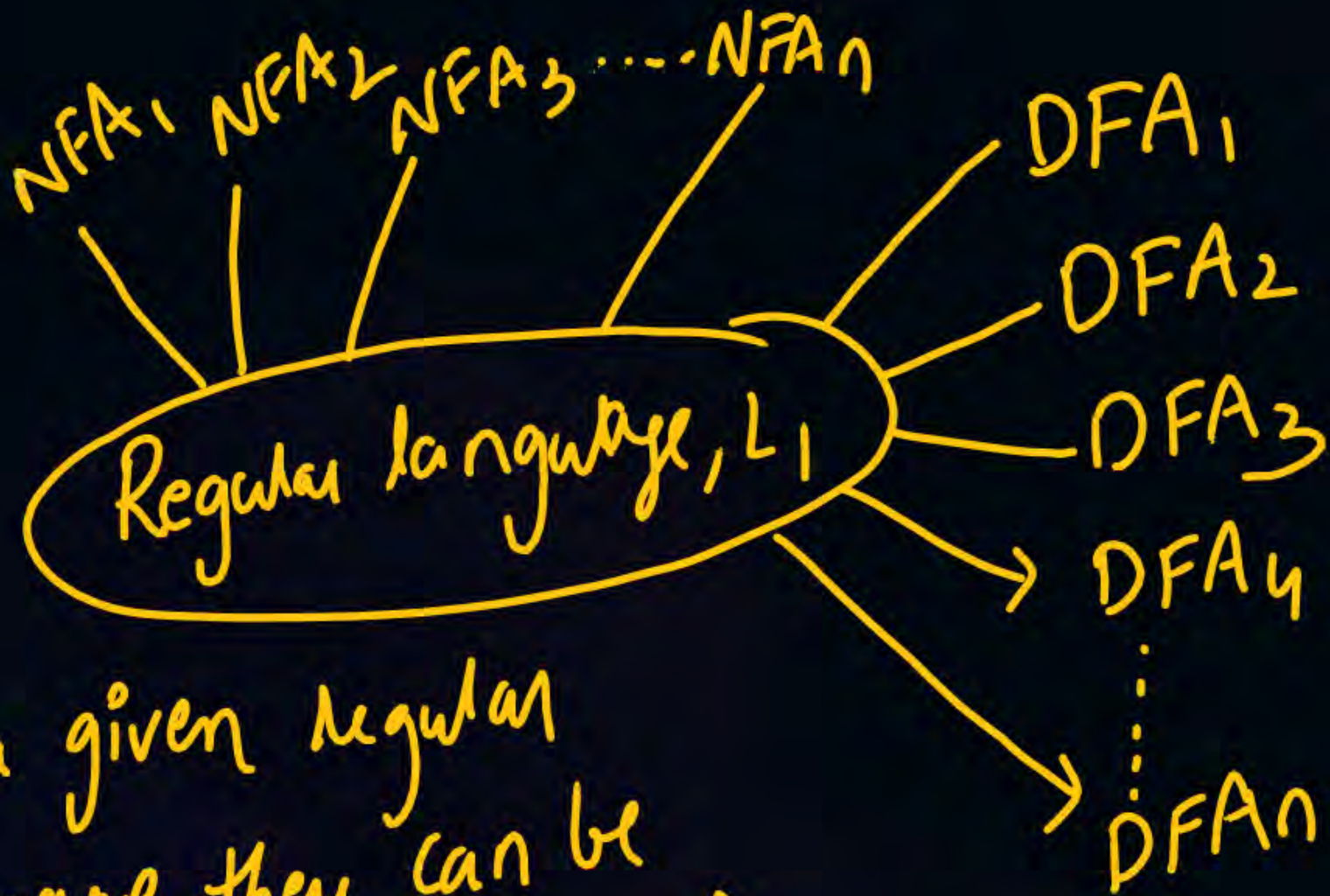
$Q, \Sigma, \delta, q_0, F$   
 $S \quad S \quad \downarrow \quad S \quad S$

$$\delta: Q \times (\Sigma \cup \{\epsilon\}) \rightarrow 2^Q$$



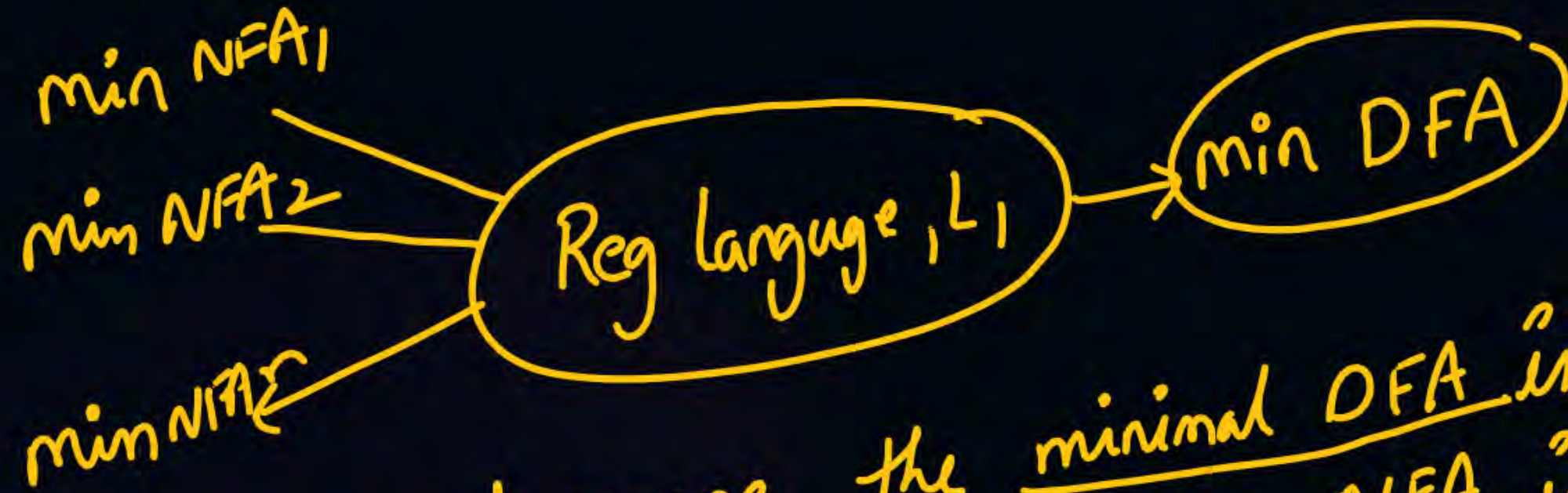
E-NFA, NFA, DFA  $\rightarrow$  all are same in power





For a given regular language, there can be many DFA's and NFA.





For a regular language, the minimal DFA is always unique  
 " NFA is not unique

" But all the min NFAs will have same no of states



## Regular expressions:- operators:-

- 1)  $+$   $\rightarrow$  union
- 2)  $\cdot$   $\rightarrow$  Concatenation
- 3)  $*$   $\rightarrow$  Kleene closure

- a)  $\phi, \epsilon, a \in \Sigma \rightarrow$  primitive regular expressions
- b) if  $r_1$  and  $r_2$  are regular expressions  
 $r_1 + r_2, r_1 \cdot r_2, r_1^*, r_2^* \rightarrow$  regular exp
- $r^*$   $\rightarrow$  Kleene closure  
 $r^+$   $\rightarrow$  positive closure

100%



$$\Sigma = \{a, b\}$$

$$RE = \underline{a} \rightarrow L = \{\underline{a}\}$$

$$RE = \underline{\epsilon} \rightarrow L = \{\underline{\epsilon}\}$$

$$RE = \phi \rightarrow L = \{\}$$

$$RE = a^* \rightarrow L = \{\epsilon, a, aa, aaa, \dots\}$$

$$RE = a^+ \Rightarrow L = \{a, aa, aaa, aaaa, \dots\}$$

$$\Sigma = \{a, b\}$$

$$\underline{(a+b)}^* \rightarrow (a+b)^0 \cup (a+b)^1 \cup (a+b)^2 \dots$$

$$\underline{(e, a, b, aa, ab, ba, bb \dots)} \downarrow \text{universal language}$$



Q.10. Let  $R = \{a^n b^m \mid n \geq 1, m \geq 0\}$ .  
 RE =  $a b^*$   $\rightarrow L = \{a, ab, abb, abbb, \dots\}$   
 a followed by any no of b's.

$\underline{a + b} \rightarrow \{a, b\}$

$\underline{a \cdot b} \rightarrow \{ab\}$

$$b^*a = \{a, ba, bba, bbba, \dots\}$$

↓  
any no of b's followed by a "a"

$$\underbrace{(b^0)}_{\epsilon}a$$

$$\epsilon \cdot a$$



$$\underline{a}^* a = \{\epsilon, a, aa, aaa, \dots\} \cdot a = \{a^1, a^2, a^3, a^4, \dots\}$$

$$= a^+ \rightarrow \text{positive closure}$$

$$\epsilon \cdot a = a \cdot \epsilon = \epsilon \cdot a \cdot \epsilon = \underline{a}$$

$$a a^* = a \{ \epsilon, a, aa, aaa, \dots \}$$

$$= \{ a, aa, aaa, \dots \} = a^+$$

$$a^* a = a a^* = a^+$$

$$\begin{aligned} \underline{a^+ + \epsilon} &= \{a, aa, aaa, aaaa, \dots\} \cup \{\epsilon\} \\ &= \{\epsilon, a, aa, aaa, \dots\} = \underline{a^*} \end{aligned}$$



$b^*ab^*$

$\Sigma = \{a, b\}$

$RE = b^*ab^* = \{ \underline{a}, ba, bab, ab, bba, \underline{a}bb, \dots \}$

all the strings containing exactly one 'a'

$$\Sigma = \{a\}$$

$$a^* = \{\epsilon, a, aa, aaa, \dots\}$$

$$(aa)^* = \{\epsilon, aa, aaaa, aaaaaa, \dots\}$$

Even number of a's



$$\Sigma = \{a\}$$

$$\frac{(aa)^* \underline{a}}{\text{Even}} \rightarrow \text{odd no of } a \text{ is}$$

$$\downarrow$$

$$(2n) + (1)$$

$$\Sigma = \{a\}$$

$$(aaa)^* \rightarrow \text{multiple of '3'}$$

$$\{\epsilon, a^3, a^6, a^9, a^{12}, \dots\}$$



$$\Sigma = \{a\}$$

$$(aaa)^* a \checkmark \rightarrow a, a^4, a^7, a^{10}, \dots$$

$$\frac{\text{no of } a\text{'s}}{(aaa)^* a} \equiv 1 \pmod{3}$$

$\downarrow$   
 congruents

$$(aaa)^*aa = a(aaa)^*a = aa(aaa)^*$$

$$a's \equiv 2 \pmod{3}$$



$$(\underline{\underline{aa}} + \underline{\underline{aaa}})^{\circledast} = \{\epsilon, \underline{\underline{aa}}, \underline{\underline{aaa}}, a\underline{\underline{aa}}, a^5, a^6, a^7, a^8\}$$

Diagram illustrating the structure of the set  $\{a^7, a^8\}$ :

- $a^7$  branches into  $\underline{\underline{aa}}$ ,  $\underline{\underline{aa}}$ , and  $\underline{\underline{aaa}}$ .
- $a^8$  branches into  $\underline{\underline{aa}}$ ,  $\underline{\underline{aa}}$ , and  $\underline{\underline{aa}}$ .

$$(\underline{\underline{aa}} + \underline{\underline{aaa}})(\underline{\underline{aa}} + \underline{\underline{aaa}})$$

$$\Sigma = \{a, b\}$$

$(a+b)^*$  → universal language  $\in, a, b, aa, ab, ba, bb, \dots$

$$\underbrace{((a+b)(a+b))^*}_{\text{even length string}} \quad ((a+b)^2)^* \quad (a+b)^{2*} \quad \underline{(a+b)^{2n}} / n \geq 0$$



11:00

$$\left( \underbrace{(a+b)(a+b)}^* \right)^* \circ \left( \underline{a} \underline{a} + \underline{b} \underline{a} + \underline{a} \underline{b} + \underline{b} \underline{b} \right)^*$$

even

$(\epsilon, aa, ba, ab)$  aaba aacb

$$\underbrace{(a+b)(a+b)}_{\text{even} + 1}^* (a+b) = \text{odd}$$



$$(a+b)(a+b)(a+b)^* \rightarrow \text{multiple of } 3$$

$6, 9, 12, \dots$

$$\underline{\underline{(a+b)(a+b)(a+b)^*}} \underline{\underline{(a+b)}} \rightarrow \underline{\underline{\cong \pmod{3}}}$$

$$\underline{\underline{a}}(a+b)^* = a \{ \epsilon, a, b, aa, ab, ba, bb, \dots \}$$

$$= \{ a, aa, ab, aaa, aab, aba, \dots \}$$

$$\underline{(a+b)^*a} = \text{ending with 'a'}$$

$$\underline{\underline{a}}(\underline{\underline{a+b}})^*\underline{\underline{b}} \rightarrow \text{starting with 'a' and ending 'b'}$$



$(a+b)^* \textcircled{ab} (a+b)^*$   
 Substring 'ab'  
 Containing 'ab'  
aa

$(a+b)^* \textcircled{a} (a+b)^*$   
 a's      a's  
 Containing 'a'  
 atleast one 'a'  
aaa

**THANK - YOU**